Lab 11

Overview of Generative AI:

Generative AI refers to systems that can create new content such as text, images, music, or even code. These systems **learn patterns** from data and use them to generate new, similar content.

Common Generative AI Models:

- **GANs (Generative Adversarial Networks)**:
  Two neural networks (Generator & Discriminator) compete against each other to produce realistic data.
- **Diffusion Models**:
  Generate data by learning to reverse a gradual process of adding noise. Very effective for **image generation** (e.g., Stable Diffusion).
- **Transformers & LLMs**:
  Language models that generate human-like text, code, and conversations (e.g., GPT, BERT, T5).

What is GenAI?

GenAI (Generative AI) is the **branch of AI focused on content creation** rather than just decision-making or classification. Examples include ChatGPT (text), DALL·E (images), and Codex (code generation).

What is LLM (Large Language Model)?

LLMs are deep learning models trained on massive datasets of human language. Capable of:

- Language understanding & generation
- Summarization
- Translation
- Sentiment analysis
- Chatbots and virtual assistants

Examples: GPT-4, PaLM, LLaMA, Falcon

What is RAG (Retrieval-Augmented Generation)?

RAG = LLM + Search

Combines language generation with real-time document retrieval. Allows an LLM to **retrieve facts** from external sources (like a database or PDF) before answering a query.

Use Case: Chatbots that pull live information from company documents.

What is FAISS?

Facebook AI Similarity Search (FAISS) is a library for **efficient similarity search** and clustering of dense vectors. It's used to **search through embeddings** quickly (e.g., finding similar documents or questions).

What is a Vector in AI?

A **vector** is a list of numbers that represents **meaningful data** (like a word, image, or sentence) in numeric form. These are often created using models like BERT, Word2Vec, or CLIP.

Example:

"apple" → [0.21, -0.34, 0.76, ..., 0.02]

These numbers represent semantic meaning.

What is a VectorDB (Vector Database)?

A **Vector Database** stores and indexes vectors (embeddings) for **fast similarity search**. Useful in GenAI apps where you want to **retrieve similar documents or questions**.

Popular examples: Pinecone, Weaviate, ChromaDB, Qdrant

What is Hugging Face?

Hugging Face is a **platform and library** for working with NLP and AI models, especially **LLMs and Transformers**. Features:

- Transformers library (BERT, GPT, T5, etc.)
- Pre-trained models and datasets
- Model training, hosting, and inference APIs

Hugging Face makes it **easy to access, fine-tune, and deploy AI models** in research and production.

LLMs and Their Applications

- Language generation (text completion, story writing)
- Virtual Assistants and Chatbots
- Summarization and Translation
- Coding assistants
- Automated customer support
- Question Answering Systems

Hugging Face Framework for Generative AI

Hugging Face offers:

- transformers – A Python library to load and use pretrained models
- datasets – Ready-to-use NLP datasets
- accelerate – For model training at scale
- spaces – To deploy your GenAI apps (like mini web apps)

Example:

```
from transformers import pipeline
chatbot = pipeline("text-generation", model="gpt2")
print(chatbot("Hello, I'm a chatbot built with Hugging Face!"))
```

Lab 11 - Tasks

Task 1; Describe the Difference Between the Following Concepts:

1. **LangChain**
2. **LLMs (**Large Language Models**)**
3. **RAG (**Retrieval-Augmented Generation**)**
4. **FAISS (**Facebook AI for Semantic Search**)**
5. **Vector**
6. **VectorDB**
7. **Generative AI (**GenAI**)**
8. **GANs (**Generative Adversarial Networks)